

Does Volatility Predict Next-Day Returns?

Martingale · Research Note 002 Author: Neil Gilani · Reproducible code: [experiment.py](#)

Abstract

We test whether a stock's recent volatility predicts the direction of its next-day return, using 2,902 trading days of SPY data (January 2015–July 2026). Pairing each day's 20-day volatility with the *following* day's return across 2,881 days, we find a correlation of **+0.035** — statistically borderline ($p \approx 0.06$) and negligibly small in practice ($R^2 \approx 0.1\%$). We conclude that recent volatility does **not** meaningfully predict the direction of the next day's move, consistent with the view that volatility measures the *size* of moves rather than their *direction*.

1. Introduction

Markets clearly go through calm stretches and stormy ones — a fact so visible that traders often assume the storminess itself carries a signal. A natural question follows: if a stock has been unusually volatile lately, does that tell us anything about where it goes *next*?

This note tests one precise version of that question. **Hypothesis (stated before running the experiment):** a day's recent volatility has *no* reliable relationship with the *direction* of the next day's return — volatility should measure how *big* moves are, not which *way* they go. A correlation near zero would support this; a clearly non-zero correlation would refute it. Fixing the hypothesis in advance is a rule of this lab's [methodology](#): it prevents inventing a story to fit whatever result appears.

2. Data & Method

Data. Daily closing prices for SPY (an ETF tracking the S&P 500) from January 2015 to July 2026 — 2,902 trading days — downloaded via [yfinance](#).

Returns. Each day's return is `today's close / yesterday's close - 1`.

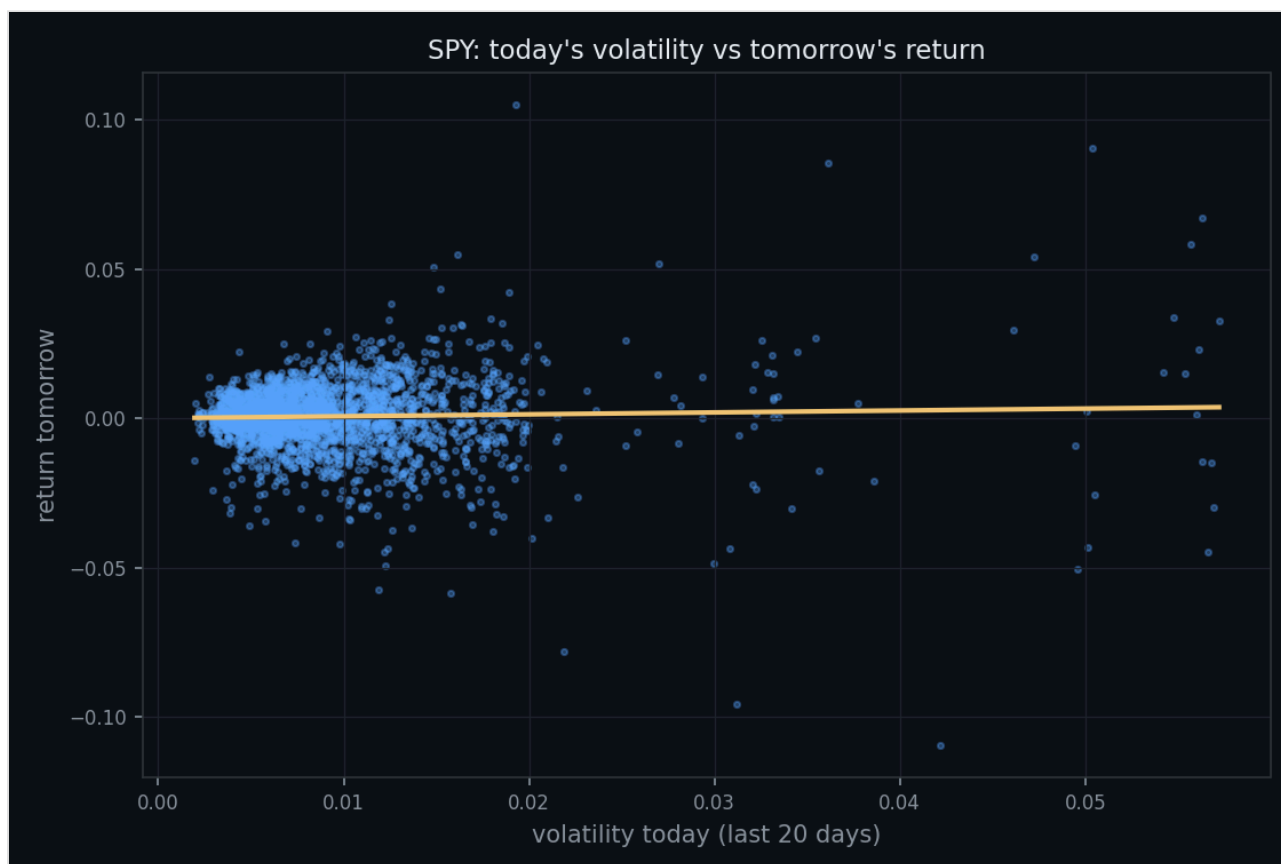
Volatility. For each day, we take the standard deviation of the trailing 20 daily returns, a standard measure of recent price "bounciness." The first 19 days lack enough history and are excluded.

The key step (no lookahead). Each day's volatility — computed only from returns *up to and including that day* — is paired with the return on the *next* day. Pairing it with the *same* day's return would let the predictor see the very move it is meant to forecast: that is lookahead bias, the error quantified in [Note 001](#). Using tomorrow's return keeps the test honest. This yields 2,881 valid (volatility, next-day return) pairs.

Metric. We report the Pearson correlation between today's volatility and tomorrow's return — a single number from -1 to $+1$, where 0 means no linear relationship.

3. Results

- Paired days analyzed: **2,881**
- Correlation (today's volatility → tomorrow's return): **+0.0347**



Across nearly 2,900 days, the correlation is **+0.035** — barely above zero. The best-fit line through the scatter is almost perfectly flat: knowing today's volatility moves our expectation of tomorrow's return by almost nothing.

4. Discussion

The result says today's volatility does **not** usefully predict the direction of tomorrow's return. Two ways to see how small +0.035 is:

- **Practical size.** Squaring the correlation gives $R^2 \approx 0.0012$ — today's volatility "explains" about **0.1%** of tomorrow's return. The other ~99.9% is unrelated to how volatile today was.
- **Is it even real?** With 2,881 data points, the noise floor for a correlation is roughly $1/\sqrt{n} \approx 0.019$. Our +0.035 sits only about **1.9 standard errors** from zero ($p \approx 0.06$), which does not clear the usual 5% significance bar. In other words, a correlation this small could plausibly be luck.

Both readings point the same way: there is, at most, a whisper of a positive relationship, and it is far too small to act on. This matches the hypothesis and echoes Note 001's broader theme — easy, obvious predictors of market direction tend to evaporate under honest testing.

5. Limitations

This study is deliberately narrow. It examines a single instrument (SPY), a single volatility window (20 days), and a single historical period; a different asset, window, or era could differ. It tests only the *direction* of the next-day return, not its *size* — volatility may well predict how *large* tomorrow's move is without predicting which *way* it goes, which would be a natural follow-up study. Pearson correlation captures only *linear* relationships, so a non-linear link could be missed. And correlation is not causation.

6. Conclusion

We asked whether recent volatility predicts the next day's return on SPY. Across 2,881 days the correlation was **+0.035** — practically and statistically indistinguishable from zero. Recent volatility does not tell you which way the market goes tomorrow; it is a measure of turbulence, not of direction.

References

- Cont, R. (2001). *Empirical properties of asset returns: stylized facts and statistical issues*.
 - Martingale, Research Note 001 — *How Backtests Lie*.
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How to cite

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